Tecniche di misura delle Temperature

misura di resistenza elettrica

$R = R(T)$

2 classi di materiali → metalli
→ semiconduttori

METALLI

$$R(T) = R_0\left[1 + a_1 T + a_2 T^2 + \dots\right]$$

$$\hookrightarrow R(T) \cong R_0\left[1 + aT\right]$$ ← metalli range temperature

$$J = e\underbrace{(\mu_p p + \mu_n n)}_{\sigma \text{ conducibilità}} E$$

densità portatori

$$\rho = \frac{1}{\sigma} = \frac{1}{\mu_p \cdot p + \mu_n \cdot n}$$

$$\mu = \frac{e}{m}\tau$$

mobilità

tempo rilassamento

* per i metalli $p \sim 0$

$$\boxed{R = \rho \cdot \frac{l}{A}}$$

Pt	-270 ~ 1000 °C	
Cu	-200 ~ 250 °C	
Ni	-200 ~ 430 C	
W	-250 ~ 1100 °C	

nome sonda

Pt100

metallo

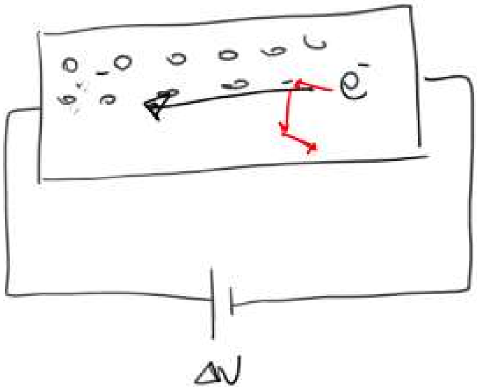
$R[\Omega]$ $T = 0$ °C

effetto Joule $P_{term} = I^2 R_x$

AUTORISCALDAMENTO

↓ soluzione

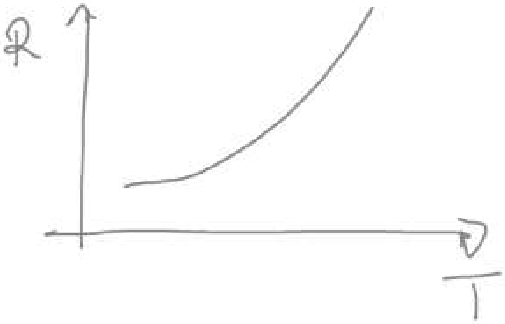
misure impulsate



$R \sim \frac{1}{\tau}$ ← tempo di rilassamento

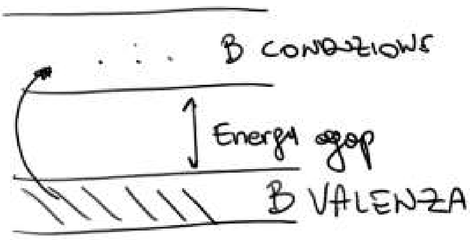
(Tempo che intercorre tra 2 urti)

$T\uparrow$ maggiori vibrazioni reticolari $\tau\downarrow$ $R\uparrow$



SEMICONDUTTORI

$R = R(T)$



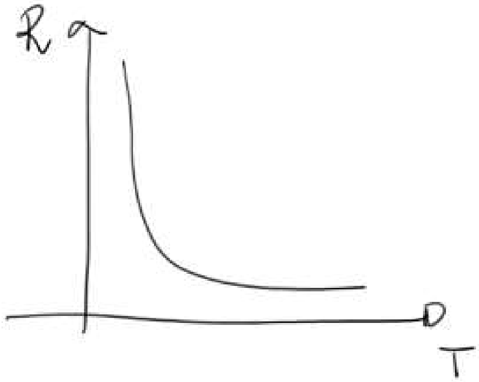
$R(T = 0K) \to \infty$

$T\uparrow$ $n\uparrow$ $\tau\downarrow$ $R\downarrow$

n ← concentrazione portatori in BC

$\propto e^{1/T}$

$\propto T$



$$R = R_0 \exp\left[\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)\right]$$

R_0: $R(T=298\,K)$

β: $\sim 4\ 10^3\,K$

$T_0 = 298\,K$

Semi C ↑ sensibilità

Metalli ↑ range di T

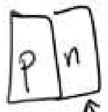
Sonde Tipicamente → Si
→ Ge [drogati]

range T $-10°C \sim 50°C$

$\frac{\partial R / \partial T}{R} = -4/5 \; 10^{-2}$ Semiconduttore

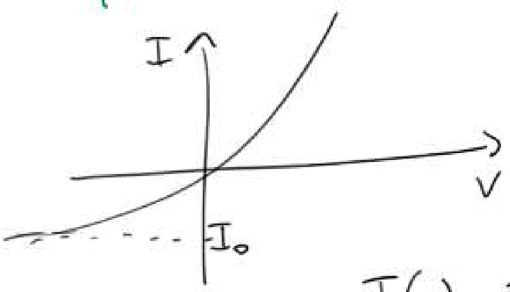
→ $Pt \sim 4 \; 10^{-3}$

SENSORI CON GIUNZIONI p/n



dropato con accettor (lacune)

dropato con donore (e^-)



$$I(V) = I_0 \left[\exp\left(\frac{qV}{k_B T}\right) - 1 \right]$$

range temperatura $-55°C \sim +155°C$

Pirometri

→ puntuali
→ mappe (telecamere)

spettro elettromagnetico misurato

VISIBILE — INFRAROSSO

(300 - 780) nm (780 nm - 1 μm)

VIOLA ROSSO

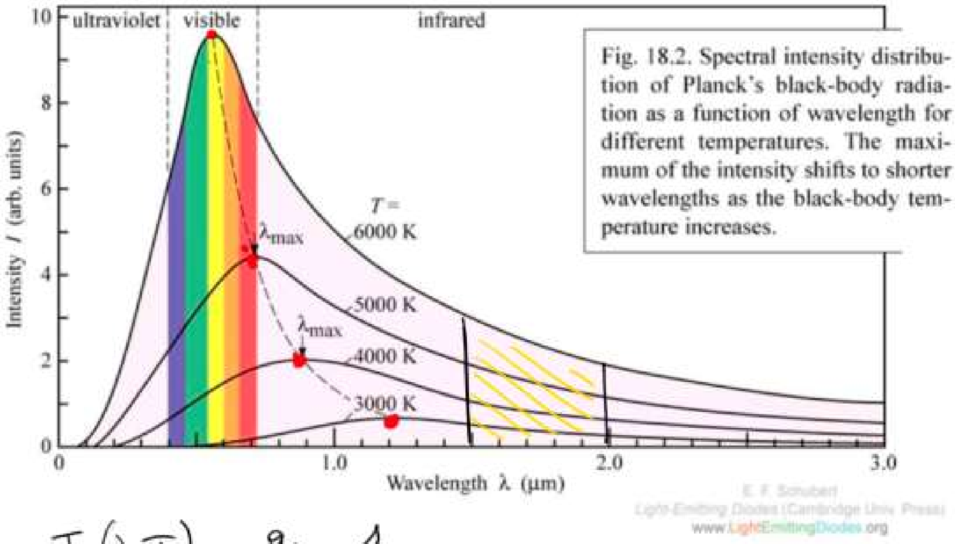


Fig. 18.2. Spectral intensity distribution of Planck's black-body radiation as a function of wavelength for different temperatures. The maximum of the intensity shifts to shorter wavelengths as the black-body temperature increases.

$$I(\lambda, T) = \frac{a}{\lambda^5} \frac{1}{e^{\frac{b}{\lambda T}} - 1} \qquad a = 2\pi \hbar c^2 \qquad b = \frac{\hbar c}{k_B}$$

Legge di Wien

$$T[K]\, \lambda_{max} = 2.897768 \cdot 10^6 \text{ nm K}$$

Legge di Stefan-Boltzmann

$$L(T) = \int_0^\infty I(\lambda, T)\, d\lambda = \sigma T^4$$

$$\sigma = \frac{2\pi^5 k_B^4}{15 c^2 \hbar^3} = 5.67 \cdot 10^{-8} \frac{J}{s\, m^2 K^4}$$

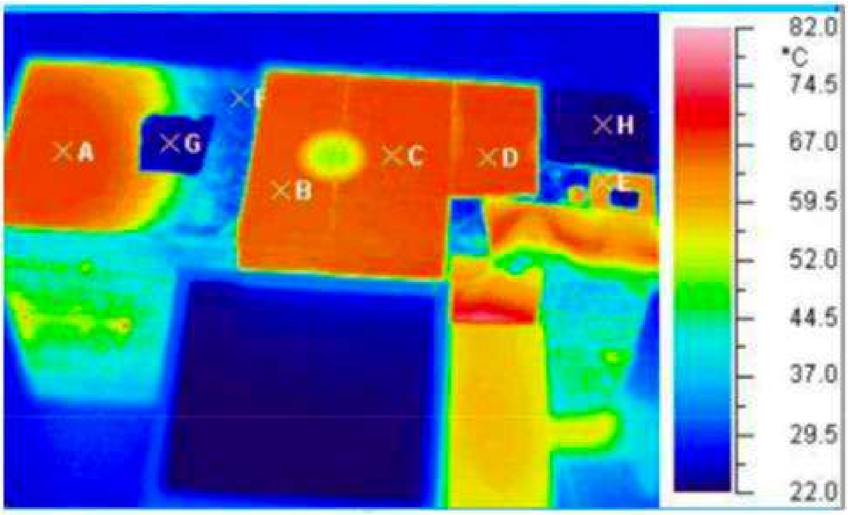
emissività

$$\varepsilon_{\lambda,T} = \frac{L(T)^{exp}}{L(T)^{\text{corpo nero}}} \qquad [\varepsilon] = 1 \qquad \varepsilon_{reale} < 1$$

$\varepsilon = \varepsilon(\lambda, T)$ $\to$ corpi grigi $\varepsilon = \varepsilon(T)$

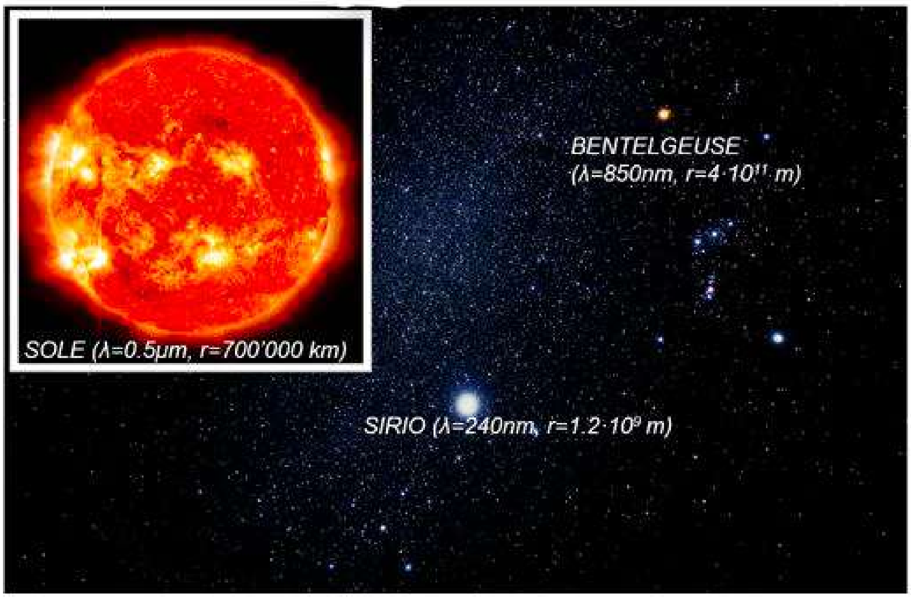
$$L(T) = \varepsilon \sigma T^4$$

caratteristica di ogni materiale

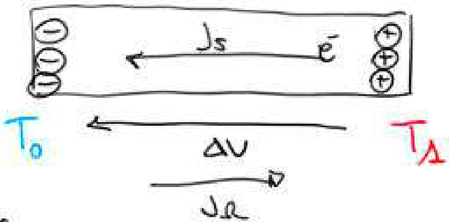


Materiali con emissività diversa, nonostante abbiano la stessa T la telecamera fornisce un'informazione errata (tarata per 1 specifica ε)

Qual è la stella più calda? ... SIRIO



Termocoppie $\longrightarrow$ effetto Seebeck



$T_1 > T_0$

diffusione di portatori dal lato caldo al lato freddo

$$J_v = -\sigma \nabla V$$

σ: conducibilità

$$\begin{cases} J_\Omega = -\sigma \nabla V \\ \quad\uparrow \\ \text{conducibilità} \\ \quad\downarrow \\ J_S = -\sigma S \nabla T \end{cases}$$

coeff Seebeck

d

$$J_{TOT} = J_S + J_\Omega = -\sigma \left[\nabla V + S \nabla T\right]$$

STATO STAZIONARIO

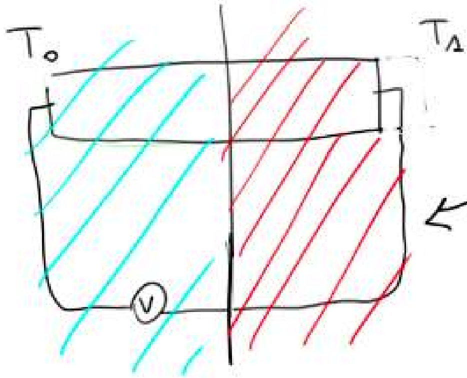
$$J_{TOT} = 0 \longrightarrow \nabla V + S\nabla T = 0 \rightsquigarrow S \equiv -\frac{\partial V}{\partial T}$$

$$[S] = \frac{V}{K}$$

$$S \sim 1 \sim 100 \frac{\mu V}{K}$$

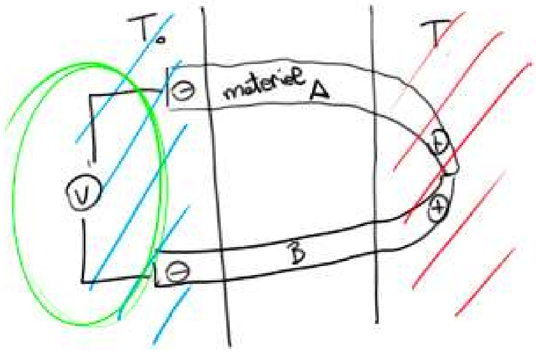
$$dV = -S\, dT$$

$$\hookrightarrow \Delta V(T_0, T_1) = -\int_{T_0}^{T_0} S(T)\, dT$$



Cavi risentono del gradiente di temperatura...

$T > T_0$



$$\Delta V_A(T) = -\int_{T_0}^{T} S_A(\tau)\, d\tau$$

$$\Delta V_B(T) = -\int_{T}^{T_0} S_B(\tau)\, d\tau$$

$$\Delta V_{AB} = \Delta V_A - \Delta V_B$$

$$\Delta V_{AB} = -\int_{T_0}^{T} S_A(\tau)\,d\tau - \int_{T}^{T_0} -S_B(\tau)\,d\tau = -\int_{T_0}^{T} \underbrace{[S_A - S_B]}_{S_{AB}}\,d\tau$$

T (°C)	ΔV (mV)
−20	−0,995
−10	−0,501
0	0
10	1,019
20	1,537

TABELLE TARATURA

Temperatura di riferimento

$T_0 = 0\,°C$

per realizzare la tabella.

In condizioni "standard" $T_0' \neq T_0 = 0\,°C$

[T_0' temperatura laboratorio]

$$\Delta V_{AB}'(T) = -\underbrace{\int_{T_0'}^{T} S_{AB}(\tau)\,d\tau}_{\text{misura sperimentale}}$$

relazione integrale e quindi soggetta alle proprietà degli integrali ...

T non ottenibile da tabelle taratura $\Delta V_{AB}' \neq \Delta V_{AB}$!

$$\underbrace{\Delta V_{AB}'(T)}_{\text{MISURATO}} = -\int_{T_0'}^{T_0} S_{AB}(\tau)\,d\tau - \underbrace{\int_{T_0}^{T} S_{AB}(\tau)\,d\tau}_{\Delta V_{AB}\ \text{TABULATO}}$$

?

$$\Delta V_{AB}(T) = \underbrace{\Delta V_{AB}'(T)}_{\text{misurato}} - \underbrace{\int_{T_0}^{T_0'} S_{AB}(\tau)\,d\tau}_{\text{ricavato (2)}}$$

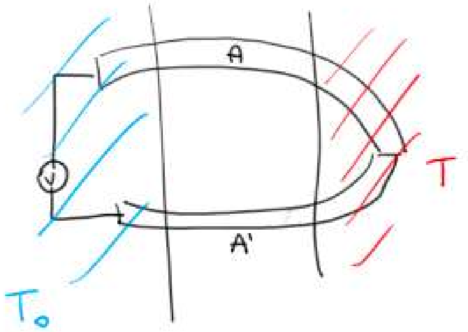
T (°C)	ΔV (mV)
(1) T_0'	(2) $\Delta V(T_0)$
T	(3) ΔV_{AB}

(1) T_0' misurata con 2° sensore

nota T_0' ricavo da tabelle $\Delta V(T_0)$

(3) ottenuto ΔV_{AB} dalle tabelle ricavo T

Termocoppie realizzate con "1 solo metallo"



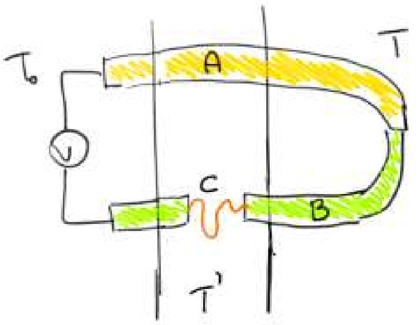
$$\Delta V_A(T) = -\int_{T_0}^{T} S_A(\tau)\, d\tau$$

$$\Delta V_{A'}(T) = -\int_{T}^{T_0} S_A(\tau)\, d\tau$$

$$\Delta V_{AA'}(T) = -\int_{T_0}^{T} (S_A - S_A)\, d\tau = 0$$

A e A' sono lo stesso materiale

Termocoppie giuntate / Temperature $T' \neq T_0 \neq T$ lungo Termocoppie



$$\Delta V_{TOT} = -\int_{T_0}^{T} S_A(\tau)\, d\tau - \left[-\int_{T}^{T'} S_B(\tau)\, d\tau - \int_{T'}^{T'} S_C(\tau)\, d\tau - \int_{T'}^{T_0} S_B(\tau)\, d\tau \right] = -\int_{T_0}^{T} (S_A - S_B)\, d\tau$$